CattleSense

ML-Based Early Detection of Cattle Diseases

MASTITIS ASSESSMENT & VETERINARY REVIEW REPORT

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Subject Cow:	[redacted] (Manali)	**Farmer / Farm:**	[redacted] (Chamara Perera)
AI Prediction:	**Mastitis** (98.5%)	**Severity Staging:**	Severe Clinical Mastitis
Priority:	**CRITICAL VETERINARY ATTENTION REQUIRED**	**Assessment Mode:**	Assisted Hybrid (Image + Biomarkers)

■■ CRITICAL VETERINARY ATTENTION REQUIRED

The CattleSense assessment identified findings associated with a potentially serious mastitis case. Prompt veterinary examination is recommended. Clinical flags: Reported udder swelling, Elevated udder warmth, Visible udder pain / kicking during milking, Abnormal milk appearance (clots/discoloration).

1. COW PROFILE & LONGITUDINAL HEALTH HISTORY

Ear Tag / ID:	COW-LK-7749	**Cow Name:**	[redacted] (Manali)
Breed:	Holstein Friesian Cross	**Age / Gender:**	4 years / Female
Lactation Count:	3	**Registration Date:**	2023-01-20
Farmer Name:	[redacted] (Chamara Perera)	**Farm Name:**	Green Meadow Dairy Farm
District / Location:	Kandy, Central Province	**Contact Phone:**	+94 77 123 4567

Longitudinal Health Trend: **Worsening** — *Severity progressed from Mild to Severe over the last 14 days.*

Date	Prediction	Severity Staging	Confidence	Decision Margin
Aug 25, 2026	Mastitis	Severe Clinical Mastitis	98.5%	Confident
Aug 18, 2026	Mastitis	Moderate Mastitis	82.0%	Confident
Aug 10, 2026	Mastitis	Mild Mastitis	65.0%	**Borderline**
Aug 01, 2026	Normal	Normal	94.0%	Confident

2. FARMER-REPORTED CLINICAL OBSERVATIONS (Q&A;)

Qualitative clinical observations reported by the farmer during assessment triage. These inform clinical severity staging and are not inserted into Model 2 numerical inputs.

Observation Question	Farmer Response	Clinical Signification
Milk Production Change	decreased	Drop in daily yield often accompanies acute inflammation
Milk Appearance / Texture	clots	Clots, flakes, or watery milk indicate secretory disruption
Milk Clotting / Flakes	yes	Visible clots or flakes resulting from casein and protein aggregation
Udder Swelling	severe	Local quarter edema from bacterial invasion / leukocyte influx
Udder Warmth / Heat	yes	Increased local vascular perfusion due to inflammatory response
Udder Pain / Tenderness	yes	Discomfort during milking or palpation
Systemic Temperature / Fever	39.8	Elevated rectal temperature (>39.2°C) indicates systemic involvement
Appetite / General Condition	poor	Anorexia / lethargy signals acute toxic or systemic mastitis

3. NUMERICAL BIOMARKERS & MODEL 2 ANALYSIS

Model 2 Active: Decision Tree Classifier (5 required milk parameters evaluated). Model Prediction: **Mastitis** (Conf: 97.5%).

Clinical Feature	Submitted Value	Availability	Reference Info
Milk Temperature	38.6 °C	**Provided**	35.0 – 37.0 °C (Normal fresh milk)
Milk pH	7.1	**Provided**	6.5 – 6.8 (Normal fresh milk)
Milk Conductivity	6.8 mS/cm	**Provided**	4.0 – 5.5 mS/cm (Normal)
Milk Yield	8.5 L/day	**Provided**	Daily milk yield in liters
Milk Clotting	1 (Clots Present)	**Provided**	0 (No clotting) / 1 (Clots/flakes)

Milk Yield Record: 8.5 L | **Source:** Auto-fetched from latest recorded Milk Log.

4. EXPLAINABLE AI — GRAD-CAM VISUAL INTERPRETATION

Research Novelty — Farmer-Guided ROI & Explainable Image-Based Screening

CattleSense incorporates farmer-guided udder region selection (ROI) combined with Grad-CAM (Gradient-weighted Class Activation Mapping) into the image-based mastitis screening workflow. Instead of presenting only a black-box prediction, the system focuses Model 1 analysis on the farmer-delineated udder and teats to minimize background noise, and produces a visual attention heatmap highlighting the regions that contributed most strongly to the ResNet50 classification. This provides transparent visual evidence for veterinary review alongside clinical and biomarker metrics.

Panel A: Original Farmer Photo   *Panel B: Selected Udder ROI*   *Panel C: Grad-CAM Heatmap*   *Panel D: Heatmap / ROI Overlay*

Figure: *Multimodal visual evidence. Panel A shows full photo; Panel B shows farmer-delineated ROI; Panels C & D display Grad-CAM activation heatmap and overlay produced by ResNet50 on cropped region.*

Model Architecture:	ResNet50 (Stage 1, frozen backbone)	**Target Conv Layer:**	conv5_block3_out (7×7×2048)
Predicted Class:	Mastitis	**Model 1 Confidence:**	99.5%
ROI Processing:	Farmer-Selected Udder ROI (Cropped)	**Model Input Dimensions:**	224 × 224 × 3 RGB (Letterbox [-1, 1])

5. MULTIMODAL HYBRID ASSESSMENT & SEVERITY STAGING

Analysis Component	Model / Methodology	Component Output	Contribution to Final Output
Image Modality	ResNet50 (Model 1)	Mastitis (99.5%)	50% (Soft-Voting Probability Fusion)
Numerical Modality	MLP Model 2	Mastitis	50% when present, 0% when omitted
Clinical Observations	7-Parameter Farmer Questionnaire	Severity: Severe Clinical Mastitis	Guides Clinical Severity & Immediate Protocol
Final Combined Assessment	**Mastitis**	**Confidence: 98.5%**	**Mode: Assisted Hybrid (Image + Biomarkers)**

6. WHAT THE FARMER SHOULD DO NOW (IMMEDIATE GUIDANCE)

1. Prompt Veterinary Consultation: Contact a qualified veterinary practitioner promptly, especially when systemic symptoms (fever, weakness, loss of appetite) or severe udder swelling are observed.
2. Strict Teat & Milking Hygiene: Ensure meticulous pre- and post-milking teat dipping using approved disinfectant solutions. Always milk suspected cows last to prevent cross-contamination.
3. Milk Isolation: Isolate and discard milk from affected quarters in accordance with farm hygiene and milk marketing regulations.
4. Clean & Dry Housing: Provide clean, dry bedding (sand, sawdust, or clean straw) to minimize environmental bacterial exposure (e.g. *E. coli*, *Streptococcus uberis*).
5. Continual Animal Monitoring: Continuously observe the cow's appetite, water intake, rectal temperature, and quarter firmness.
6. Present This Report: Hand this structured case document to the attending veterinarian upon their arrival.

IMPORTANT SAFETY DIRECTIVE: Do not administer veterinary antibiotics or prescription drugs based solely on this automated AI screening. All medical and antimicrobial treatment decisions must be made by a licensed veterinarian following clinical examination and, where appropriate, diagnostic testing (e.g. Milk Culture / CMT).

7. VETERINARY REVIEW & CLINICAL HANDOVER

This section summarizes the AI decision-support findings for veterinary review and provides a structured clinical handover record. The attending veterinarian independently evaluates the animal and completes the fields below.

Attending Veterinarian:	____
License / Registration No:	____
Physical Examination Findings:	____
Diagnostic Tests Ordered (CMT / Culture):	____
Definitive Diagnosis:	____
Prescribed Treatment / Management Plan:	____
Date & Signature:	[redacted] / Date: ___________________ [redacted] / Sig: ______________________

8. AI ASSESSMENT NOTICE & VETERINARY REFERENCES

IMPORTANT AI ASSESSMENT NOTICE: CattleSense provides an AI-assisted screening and decision-support assessment. The prediction is generated from image features, numerical biomarkers, and farmer-reported triage inputs. It does NOT replace physical examination, palpation, somatic cell verification, laboratory culture, or licensed veterinary diagnosis. Grad-CAM visual heatmaps represent model predictive attention and must not be interpreted as anatomical lesion proof.

Authoritative Veterinary References:

1. **Merck Veterinary Manual:** *Mastitis in Cattle*. Ken Leslie, DVM, MSc; Christopher S. Peters, DVM. Available at: https://www.merckvetmanual.com/reproductive-system/mastitis-in-large-animals/mastitis-in-cattle
2. **Merck Veterinary Manual:** *Overview of Mastitis in Large Animals*. Available at: https://www.merckvetmanual.com/reproductive-system/mastitis-in-large-animals/overview-of-mastitis-in-large-animals
3. **National Mastitis Council (NMC):** *Current Concepts of Bovine Mastitis* (5th Edition). Laboratory Handbook on Bovine Mastitis.